WORKED SOLUTION

STEP II 2000, Question 6

Pure • Half-Angle Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q6

The key idea

The substitution $t = \tan\frac{1}{2}\theta$ turns any rational function of $\sin\theta$ and $\cos\theta$ into a rational function of t, and here the result is 2 over a quadratic. Completing the square gives an arctangent, and the third identity in the first part is exactly what you need to simplify the arctangent at the upper limit. The second integral then follows from the first by symmetry, without any new integration.

Solution

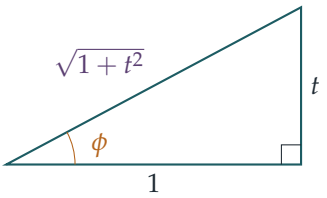
The three identities

Write $\phi = \frac{1}{2}\theta$, so $t = \tan\phi$ and $1 + t^2 = \sec^2\phi$. By the double-angle formulae,

$$\sin\theta = 2\sin\phi\cos\phi = 2\tan\phi\cos^2\phi = \frac{2\tan\phi}{\sec^2\phi} = \frac{2t}{1+t^2},$$

$$\cos\theta = \cos^2\phi - \sin^2\phi = \cos^2\phi\,(1 - \tan^2\phi) = \frac{1-t^2}{1+t^2}.$$

(These hold whenever $\tan\frac{1}{2}\theta$ is defined, that is, $\cos\frac{1}{2}\theta \neq 0$.) The right-angled triangle below, with legs 1 and t and angle ϕ, is a good way to remember them.



For the third, when $\sin\theta \neq 0$, use $1 + \cos\theta = 2\cos^2\phi$ and $\sin\theta = 2\sin\phi\cos\phi$:

$$\frac{1+\cos\theta}{\sin\theta} = \frac{2\cos^2\phi}{2\sin\phi\cos\phi} = \cot\phi = \tan(\tfrac{1}{2}\pi - \phi) = \tan(\tfrac{1}{2}\pi - \tfrac{1}{2}\theta),$$

since $\cot\phi = \tan(\frac{1}{2}\pi - \phi)$ for complementary angles. All three identities are shown.

The first integral

With $t = \tan\frac{1}{2}\theta$ we have $\theta = 2\arctan t$, so

$$\mathrm{d}\theta = \frac{2}{1+t^2}\,\mathrm{d}t.$$

When $\theta = 0$, $t = 0$; when $\theta = \frac{1}{2}\pi$, $t = \tan\frac{1}{4}\pi = 1$. Write $c = \cos\alpha$ and $s = \sin\alpha$; both are positive since $0 < \alpha < \frac{1}{2}\pi$. Then

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + c\sin\theta} = \int_0^1 \frac{1}{1 + \dfrac{2ct}{1+t^2}} \cdot \frac{2}{1+t^2}\,\mathrm{d}t = \int_0^1 \frac{2}{t^2 + 2ct + 1}\,\mathrm{d}t.$$

Completing the square, and using $1 - c^2 = s^2$,

$$t^2 + 2ct + 1 = (t+c)^2 + s^2.$$

Since $s > 0$, the standard integral $\int \frac{\mathrm{d}u}{u^2+s^2} = \frac{1}{s}\arctan\frac{u}{s}$ gives

$$\int_0^1 \frac{2}{(t+c)^2+s^2}\,\mathrm{d}t = \frac{2}{s}\Big[\arctan\frac{t+c}{s}\Big]_0^1 = \frac{2}{s}\left(\arctan\frac{1+\cos\alpha}{\sin\alpha} - \arctan\frac{\cos\alpha}{\sin\alpha}\right).$$

Ask yourself How do we simplify $\arctan\dfrac{1+\cos\alpha}{\sin\alpha}$?

This is where the first part earns its keep. By the third identity (with α in place of θ), $\frac{1+\cos\alpha}{\sin\alpha} = \tan(\frac{1}{2}\pi - \frac{1}{2}\alpha)$. But we must check the angle lies in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, the range of arctan, before we can cancel arctan against tan. Since $0 < \alpha < \frac{1}{2}\pi$, we have $\frac{1}{4}\pi < \frac{1}{2}\pi - \frac{1}{2}\alpha < \frac{1}{2}\pi$, so indeed

$$\arctan\frac{1+\cos\alpha}{\sin\alpha} = \tfrac{1}{2}\pi - \tfrac{1}{2}\alpha.$$

Similarly $\frac{\cos\alpha}{\sin\alpha} = \cot\alpha = \tan(\frac{1}{2}\pi - \alpha)$ with $0 < \frac{1}{2}\pi - \alpha < \frac{1}{2}\pi$, so $\arctan\frac{\cos\alpha}{\sin\alpha} = \frac{1}{2}\pi - \alpha$. Therefore

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + \cos\alpha\sin\theta} = \frac{2}{\sin\alpha}\Big((\tfrac{1}{2}\pi - \tfrac{1}{2}\alpha) - (\tfrac{1}{2}\pi - \alpha)\Big) = \frac{2}{\sin\alpha}\cdot\frac{\alpha}{2} = \frac{\alpha}{\sin\alpha},$$

as required.

The second integral

Substitute $\theta = \frac{1}{2}\pi - u$. Then $\cos\theta = \sin u$, $\mathrm{d}\theta = -\mathrm{d}u$, and the limits $0, \frac{1}{2}\pi$ swap, which cancels the minus sign:

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + \sin\alpha\cos\theta} = \int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}u}{1 + \sin\alpha\sin u}.$$

Now let $\beta = \frac{1}{2}\pi - \alpha$. Then $\sin\alpha = \cos\beta$, and $0 < \beta < \frac{1}{2}\pi$, so the result just proved applies with β in place of α:

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}u}{1+\cos\beta\sin u} = \frac{\beta}{\sin\beta}.$$

Since $\sin\beta = \cos\alpha$, we conclude that, for $0 < \alpha < \frac{1}{2}\pi$,

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1+\sin\alpha\cos\theta} = \frac{\frac{1}{2}\pi - \alpha}{\cos\alpha}.$$

Checks

- As $\alpha \to 0$, $\cos\alpha \to 1$ and the first integral should tend to $\int_0^{\pi/2} \frac{\mathrm{d}\theta}{1+\sin\theta} = 1$ (substitute and you get $\int_0^1 \frac{2}{(1+t)^2}\,\mathrm{d}t = 1$); and indeed $\alpha/\sin\alpha \to 1$.
- As $\alpha \to \frac{1}{2}\pi$ the integrand tends to 1, so the first integral should tend to $\frac{1}{2}\pi$, which matches $\frac{\pi/2}{1}$. The second integral likewise tends to 1 as $\alpha \to \frac{1}{2}\pi$, consistent with $\frac{\frac{1}{2}\pi-\alpha}{\cos\alpha} \to 1$.
- Numerically, at $\alpha = \frac{1}{4}\pi$ Simpson's rule gives 1.110721 for the first integral, matching $\frac{\pi\sqrt{2}}{4} \approx 1.110721$; at $\alpha = \frac{1}{3}\pi$ the second integral comes out as $1.047198 \approx \frac{1}{3}\pi$, as the formula predicts.

Common slips

- Forgetting the factor $\frac{2}{1+t^2}$ from $\mathrm{d}\theta$, or the new limits 0 and 1.
- Writing $\arctan(\tan x) = x$ without checking that x lies in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$; here the restriction $0 < \alpha < \frac{1}{2}\pi$ is exactly what makes it valid.
- Redoing the whole integration for the second integral, rather than using $\theta \mapsto \frac{1}{2}\pi - \theta$ and $\alpha \mapsto \frac{1}{2}\pi - \alpha$.

Beyond STEP

The $t = \tan\frac{1}{2}\theta$ substitution (sometimes called the Weierstrass substitution) works because the point $(\cos\theta, \sin\theta) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$ is a rational parametrisation of the unit circle; taking rational t gives every Pythagorean triple. At university, integrals of the form $\int_0^{2\pi} R(\cos\theta, \sin\theta)\,\mathrm{d}\theta$ are usually done instead by setting $z = e^{i\theta}$ and using the residue theorem, which turns the calculation into finding poles inside the unit circle. Needham's *Visual Complex Analysis* is a beautiful introduction to that idea.

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